Uniform rank bounds for finite-time propagation From independent products to all coupling chaoses

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Abstract

We turn the established rank-product covariance formula into a positive Markov chain. Its first two conditional Casimir moments give a rank-deficit bound uniform in the number of factors, not merely at fixed polynomial degree. Gaussian tensor symmetrization transfers the bound to the highest coupling chaos. A one-step version and Gaussian creation/annihilation operators then control the rank–chaos deficit of the full ordinary propagator, including all lower contractions, uniformly on compact real-time intervals. A finite-parameter covariance-filter estimate follows. We also identify an exact charged-probe four-point interface and a positive regression remainder sufficient for the actual memory-field bound. Connecting the rank–chaos estimate to the charged force remains the next step.

1 State, model and the all-degree statement

Use standard CAR, $\hbar = 1$, string time, and the overlapping complex Gaussian model $H = \sum_{I,J} J_{IJ} c_I^\dagger c_J$, with

$$\mathbb{E} J_{IJ} J_{KL} = w_N \delta_{IL} \delta_{JK}, \quad w_N = 4^p \frac{J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2, \qquad \lambda_N = p^2/N. \tag{1}$$

$J = J^\dagger$; all charge sectors use the same coupling. At charge m, $\mathrm{tr}_m = \mathrm{Tr}\,/\binom{N}{m}$ and $\|X\|_{2,m}^2 = \mathbb{E}\,\mathrm{tr}_m\, X^\dagger X$. Put

$$C = C_m = w_N \binom{m}{p} \binom{N-m+p}{p}, \qquad \Phi(X) = \mathbb{E}_{\widetilde{H}} \widetilde{H} X \widetilde{H}. \tag{2}$$

Here $\widetilde{H}$ is an independent copy; Φ is a deterministic linear map. The neutral operator algebra has multiplicity-free $U(N)$ ranks $0 \le r \le R_m = \min(m, N-m)$, with orthogonal projectors P_r. These are representation ranks, not matrix ranks.

For independent copies $H_1, \dots, H_k$ in this sector define

$$\pi_k(r) = C^{-k} \mathbb{E}\,\mathrm{tr}_m\, |P_r(H_1 \cdots H_k)|^2, \qquad D_k(r) = kp - r. \tag{3}$$

For $N \ge 8$, $1 \le p \le N/8$, $N/4 \le m \le 3N/4$, and every integer $k \ge 1$, we prove

$$\boxed{\mathbb{E}_{\pi_k} D_k^2 \le A(\lambda_N k^2 + \lambda_N^2 k^4), \qquad A = 2 \cdot 10^6.} \tag{4}$$

The constant is deliberately unoptimized. The result is a finite parameter bound. For double scaling take even $N \to \infty$, $\lambda_N \to \lambda > 0$ in a positive compact interval. The full $\beta = \mu = 0$ state uses $b_m = 2^{-N}\binom{N}{m}$.

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2 An exact rank-convolution Markov chain

The existing covariance and Racah interfaces [1, 2] give

$$\Phi(P_sX) = \kappa_s P_s X, \qquad \kappa_s = w_N \binom{m-s}{p}\binom{N-m+p-s}{p}, \tag{5}$$

$$\chi_r(s) = {}_4F_3\left(\begin{matrix}-r, r-N-1, -s, s-N-1\\ -m, m-N, -N-1\end{matrix};1\right), \quad \mu_s = \kappa_s/C. \tag{6}$$

Invalid binomials are zero. Let $d = \binom{N}{m}$ and $d_r = \binom{N}{r}^2 - \binom{N}{r-1}^2$. Covariance composition gives

$$\sum_r \pi_k(r)\chi_r(s) = \mu_s^k, \qquad \pi_k(r) = \frac{d_r}{d^2}\sum_s d_s\chi_r(s)\mu_s^k. \tag{7}$$

Positivity is not inferred from the signs of this sum. Decompose $H = \sum_a H_a$ into independent isotropic rank components and let $\omega_a = C^{-1}\mathbb{E}\,\mathrm{tr}_m\, H_a^2$. If $\pi_{ra}(t)$ is the known positive output law of two independent normalized pure-rank inputs, the transition kernel is

$$P(r,t) = \sum_a \omega_a \pi_{ra}(t), \qquad \pi_{k+1}(t) = \sum_r \pi_k(r)P(r,t), \qquad \pi_0 = \delta_0. \tag{8}$$

Indeed the sesquilinear coefficient covariance of $H_1\cdots H_k$ is $U(N)$ invariant and therefore scalar on each neutral rank. No Hermiticity of that product is required. Its covariance map is Φ^k, and the invariant Choi decomposition gives (7). Multiplication implies $r \le kp$ on its support, hence $D_k \ge 0$ even when $kp > R_m$.

Write $x_r = r(N+1-r)$, $X_j = x_{r_j}$ and $n = N-m$. Define

$$e = mn(N+1)/N, \qquad u = \mu_1 = \frac{(m-p)n}{m(n+p)}, \tag{9}$$

$$v = \mu_2 = u\frac{(m-p-1)(n-1)}{(m-1)(n+p-1)}, \qquad c = e(1-u) = \frac{pn(N+1)}{n+p}, \tag{10}$$

$$b = \frac{(N-1)(N-2)}{m(m-1)n(n-1)N(N+1)}, \qquad d_2 = \frac{2(N-1)}{Ne} + bN. \tag{11}$$

The first two characters are $\chi_r(1) = 1 - x_r/e$ and $\chi_r(2) = 1 - d_2x_r + bx_r^2$. Applying the transition to them gives

$$\mathbb{E}[X_{j+1} \mid X_j = x] = ux + c, \tag{12}$$

$$\mathrm{Var}(X_{j+1} \mid X_j = x) = (v-u^2)x^2 + Bx + V_1, \tag{13}$$

$$B = \frac{d_2}{b}(u-v) - 2uc, \qquad V_1 = \frac{p^2n^2(N+1)(m-p+1)}{(n+p)^2(n+p-1)}. \tag{14}$$

In particular $\mathbb{E}X_k = e(1-u^k)$ and $\mathbb{E}X_k^2 = (v^k - 1 + d_2\mathbb{E}X_k)/b$, exactly at finite N.

3 Uniform control without large-order cancellations

The sign $0 \le v \le u^2 \le 1$ follows by comparing the two factors in v/u to those in u. Direct simplification also gives

$$B = \frac{up\{n(m^2+n^2-2) - p(m^2+2mn-n^2-2n)\}}{(m-1)(n+p)(n+p-1)}. \tag{15}$$

Under the central-sector assumptions of (4), bound the numerator termwise and use $m-1, n+p-1 \ge N/8$. For example, the following conservative bounds all hold:

$$|B| \le 102400p, \quad V_1 \le 512p^2N, \quad c \le 8pN, \quad 1-u \le 16p/N. \tag{16}$$

The total-variance recurrence, with $M_j = \mathbb{E}X_j$, is

$$\operatorname{Var} X_{j+1} = v \operatorname{Var} X_j + (v-u^2)M_j^2 + BM_j + V_1. \tag{17}$$

Since $M_j \le jc$, discarding the nonpositive quadratic term and summing (16) yields, for every k,

$$\operatorname{Var} X_k \le 410112\, k^2p^2N. \tag{18}$$

If $t = kp \le N/8$, use

$$x_t = kx_p - k(k-1)p^2, \qquad x_p - c = \frac{p^2(m-p+1)}{n+p}, \tag{19}$$

$$kc - M_k = c\sum_{j<k}(1-u^j) \qquad \le \tfrac{1}{2}k(k-1)c(1-u). \tag{20}$$

Together with support $r \le t$, these imply $0 \le x_t - M_k \le 73k^2p^2$. Furthermore $x_t - x_r = (t-r)(N+1-t-r) \ge (N/2)(t-r)$. Thus

$$\mathbb{E}D_k^2 \le \frac{4}{N^2}\left[\operatorname{Var} X_k + (x_t - M_k)^2\right] \le 1640448\lambda_N k^2 + 21316\lambda_N^2 k^4. \tag{21}$$

For $kp > N/8$, do not use monotonicity of x_{kp}: simply $D_k^2 \le k^2p^2 \le 64\lambda_N^2k^4$. This proves (4) uniformly over all degrees.

At half filling and fixed k, the exact formulas also give

$$\mathbb{E}D_k \to \lambda k^2, \qquad \operatorname{Var} D_k \to \lambda k^2, \qquad \mathbb{E}D_k^2 \to \lambda^2k^4 + \lambda k^2. \tag{22}$$

For the mean, expand the exact $e(1-u^k)$ at fixed k. For variance, $B/p \to 2$, $V_1/N^2 \to \lambda$, $(v-u^2)p^2 \to 0$, and $M_j/(Np) \to j$ turn the recurrence into increments $\lambda(2j+1)$. Finally $(x_{kp} - X_k)/N = D_k[1 + O(kp/N)]$ uniformly on its support. These two moments match a Poisson law, but do not prove that law.

4 Highest chaos, Wick series and a covariance-filter bound

Now return to the same coupling in every factor. Let $Z_k = \mathsf{W}_k(H^k)$, where W_k is the orthogonal projection onto degree-k Gaussian coupling chaos. In real Gaussian coordinates, let T be the ordered k-linear coefficient tensor of the product. For any deterministic rank weight w, the Gaussian isometry gives

$$\|wZ_k\|_{2,m}^2 = k!\|\operatorname{Sym}(wT)\|^2 \le k!\|wT\|^2 = k!C^k\mathbb{E}_{\pi_k}|w(r)|^2. \tag{23}$$

Symmetrization is an orthogonal projection on the input tensor; it commutes with the output rank filter. There is only one $k!$.

Let $\mathcal{D}_k = \sum_r (kp - r)P_r$. Distinct coupling chaoses are orthogonal. Therefore the series

$$\mathcal{D}(s) = \sum_{k\ge 0} \frac{(is)^k}{k!}\mathcal{D}_k Z_k \tag{24}$$

is entire in annealed L^2. With $x = C|s|^2$, its squared norm obeys

$$\|\mathcal{D}(s)\|_{2,m}^2 \le Ae^x\left[\lambda_N(x^2+x) + \lambda_N^2(x^4+6x^3+7x^2+x)\right]. \tag{25}$$

This is an all-degree sum, using the exact exponential generating functions for k^2 and k^4, not a time truncation. For full charge summation use the previously established $C_m \le M$ uniformly in m, N. Outside the central region, $D_k \le kp$ and (23) give the total tail bound

$$2p^2e^{-N/8}e^x(x^2+x), \qquad x = M|s|^2. \tag{26}$$

It vanishes on compact time sets. Inactive sectors are zero.

In the same central sectors there is a finite-parameter filter consequence. Extend the exact covariance eigenvalue to $z \ge 0$ by

$$\mu(z) = \prod_{j=0}^{p-1}\left(1 - \frac{z}{m-j}\right)_+\left(1 - \frac{z}{N-m+p-j}\right)_+. \tag{27}$$

This agrees with μ_r at every allowed integer rank. The product is globally Lipschitz with constant at most $p/(m-p+1) + p/(N-m+1) \le 12p/N$. Consequently

$$|p(\mu_r - \mu(kp))| \le 12\lambda_N D_k(r), \tag{28}$$

$$\|p(\Phi/C - \mu(kp))Z_k\|_{2,m}^2 \le 144\lambda_N^2 k!C^k\mathbb{E}_{\pi_k}D_k^2. \tag{29}$$

Its corresponding Wick series is bounded by $144\lambda_N^2$ times (25). Keeping $\mu(kp)$ exact avoids amplifying the finite-parameter error of an early replacement by q^k. Neither Z_k nor this comparison term is asserted to commute with H.

5 A one-step bound controls creation and annihilation

The all-degree estimate can be extended beyond the Wick series. For one rank transition $r \to t$ with $r \le kp$, put $\delta = p + r - t$. Multiplication by a rank-a tensor has $|t - r| \le a \le p$, hence $0 \le \delta \le 2p$. For completeness, the lower triangle follows by cyclic trace: if $r > t + a$, then $\langle C_t, A_r B_a \rangle = \langle C_t B_a^\dagger, A_r \rangle = 0$ since the former product has rank at most $t + a$. No product multiplicity assumption is needed.

If $(k+1)p \le N/8$, (12)–(16) give

$$0 \le x_{r+p} - \mathbb{E}_r x_t \le 32(k+1)p^2, \tag{30}$$

$$\mathrm{Var}_r\, x_t \le 205312(k+1)p^2 N. \tag{31}$$

For example the mean difference equals $(1-u)x_r + [p(N+1) - c] - 2rp - p^2$, and $p(N+1) - c = p^2(N+1)/(n+p)$. Using $\delta \le 2(x_{r+p} - x_t)/N$ gives a bound $821248\lambda_N(k+1) + 4096\lambda_N^2(k+1)^2$. If $(k+1)p > N/8$, use $\delta^2 \le 4p^2 \le 256\lambda_N^2(k+1)^2$. Thus, with $A_1 = 10^6$, uniformly in all $k \ge 0$,

$$\sup_{r \le kp} \mathbb{E}_r \delta^2 \le A_1[\lambda_N(k+1) + \lambda_N^2(k+1)^2]. \tag{32}$$

Represent $H = \sum_\alpha g_\alpha B_\alpha$ with real standard Gaussian coordinates, Hermitian B_α, and $\sum_\alpha B_\alpha^2 = CI$. Work in Gaussian L^2 tensored with the normalized matrix Hilbert space, restricted to $U(N)$-equivariant functions. Let $\mathcal{N}$ count Gaussian chaos and $\mathcal{R} = \sum_r rP_r$ act on the matrix output. The operator

$$\mathfrak{D} = p\mathcal{N} - \mathcal{R} \tag{33}$$

is nonnegative here: degree-k equivariant coefficients have output rank at most kp. Coupling tensors are built from ranks at most p. Left multiplication by H is $\mathcal{T} = \mathcal{C} + \mathcal{C}^\dagger$, with $\mathcal{C} = \sum_\alpha a_\alpha^\dagger L_{B_\alpha}$.

The deterministic operator $\sum_\alpha L_{B_\alpha} P_t L_{B_\alpha}$ is $U(N)$ equivariant, so its cross terms between different input ranks vanish exactly; on rank r its scalar is $CP(r,t)$. Apply this to the unsymmetrized creation tensor, then use contraction of input symmetrization:

$$\|[\mathfrak{D}, \mathcal{C}]\psi_k\|^2 \le (k+1)C \sup_{r \le kp} \mathbb{E}_r \delta^2 \, \|\psi_k\|^2. \tag{34}$$

The equivariant subspace is reduced by creation, annihilation, number and rank. Thus the annihilation block is minus the adjoint of this block. Adding the two adjacent blocks gives

$$\|[\mathfrak{D}, \mathcal{T}]\psi\|^2 \le 4A_1 C(\lambda_N + \lambda_N^2)\|(\mathcal{N}+1)^{3/2}\psi\|^2. \tag{35}$$

6 The full ordinary propagator, with every chaos retained

Let $\mathcal{F}(s) = e^{isH}$ and $s \in \mathbb{R}$. Its Gaussian derivative tensor has a dimension-free bound. For any ordered Duhamel integrand

$$W_{\boldsymbol{\alpha}} = U_0 B_{\alpha_1} U_1 \cdots B_{\alpha_\ell} U_\ell, \qquad U_j \text{ unitary}, \tag{36}$$

the exact identity $\sum_{\boldsymbol{\alpha}} \operatorname{tr}_m W_{\boldsymbol{\alpha}}^\dagger W_{\boldsymbol{\alpha}} = C^\ell$ follows by successive summation using $\sum B_\alpha^2 = CI$ and cyclic trace. The $U_j = e^{it_j H}$ are allowed to depend on the sample. They do not depend on the summed derivative indices.

Minkowski in the derivative-index ℓ^2 and matrix Hilbert space, followed by the $\ell!$ permutations and simplex integration, therefore gives, already samplewise,

$$\sum_{\boldsymbol{\alpha}} \operatorname{tr}_m |\partial_{\boldsymbol{\alpha}} \mathcal{F}(s)|^2 \le C^\ell |s|^{2\ell}. \tag{37}$$

The factorial cancels the simplex volume. Gaussian chaos isometry then implies, with $z = Cs^2$ and $\mathcal{F}_k = \mathsf{W}_k \mathcal{F}$,

$$\sum_k (k)_\ell \|\mathcal{F}_k(s)\|^2 \le z^\ell, \tag{38}$$

$$\|(\mathcal{N}+1)^{3/2} \mathcal{F}(s)\|^2 \le 1 + 7z + 6z^2 + z^3. \tag{39}$$

Here $(k)_\ell = k(k-1)\cdots(k-\ell+1)$. In fact summing the nonnegative factorial moments yields $\sum_k \zeta^k \|\mathcal{F}_k(s)\|^2 \le e^{z(\zeta-1)}$ for $\zeta \ge 1$. This is a bound on the complete chaos distribution.

Since $\mathfrak{D} I = 0$, the commutator Duhamel identity is

$$\mathfrak{D}\mathcal{F}(s) = i \int_0^s e^{i(s-t)\mathcal{T}} [\mathfrak{D}, \mathcal{T}] \mathcal{F}(t)\, dt. \tag{40}$$

At finite N, all Gaussian derivatives of $\mathcal{F}$ are bounded; the number operator adds only polynomial growth in Gaussian coordinates. The identity follows on the common Hermite core and its Gaussian Sobolev closure. The left evolution is unitary for real time. Equations (35)–(39) prove

$$\boxed{\|\mathfrak{D} e^{isH}\|_{2,m}^2 \le 4A_1 C(\lambda_N + \lambda_N^2) s^2 (1 + 7z + 6z^2 + z^3).} \tag{41}$$

For full charge summation, $0 \le \mathfrak{D} \le p\mathcal{N}$ on equivariant functions bounds the exterior-sector contribution by $2p^2 e^{-N/8}(z^2 + z)$, now with $z = Ms^2$. Thus the actual ordinary propagator has a uniformly bounded rank–chaos deficit on compact real-time intervals, including all its lower contractions.

The Lipschitz argument also gives in central sectors

$$\|p[\Phi/C - \mu(p\mathcal{N})] e^{isH}\|_2 \le 12\lambda_N \|\mathfrak{D} e^{isH}\|_2. \tag{42}$$

The right comparison acts on Gaussian chaos; it is not asserted to be a samplewise conserved function of H.

7 The actual finite-time target and a positive interface

For the charge-removal dilation, let $h = H_m$, $K = 1_N \otimes H_{m-1}$, $V\psi = m^{-1/2}\sum_i |i\rangle c_i \psi$, $a = (1-p/m)h$, and $R = KV - Va$. Write $F(s) = V^\dagger e^{iKs}V$ and $\Lambda(s) = R^\dagger e^{iKs}R$. The preceding stage showed that the actual pinched memory field differs in L^2 by $o(1)$ from

$$X_0(s) = (1-\mathsf{P}_h)me^{-ihs/2}\Lambda(s)e^{-ihs/2}, \qquad \mathsf{P}_h X = \sum_e P_e X P_e. \tag{43}$$

Its uniform second moment is still the target.

Independent circular Gaussian vectors ξ, η, independent of J, give linear charged probes $M_\xi = \sum_i \xi_i c_i$ with $\mathbb{E}_\xi \bar{\xi}_i \xi_j = \delta_{ij}$. Exactly at finite N,

$$F(s) = \frac{1}{m}\mathbb{E}_\xi M_\xi^\dagger e^{iH_{m-1}s} M_\xi. \tag{44}$$

Two F insertions are therefore a four-point function with two independent probe colors. Define

$$B_a(t_1,t_2) = e^{iat_1}F(t_2-t_1)e^{-iat_2}. \tag{45}$$

Direct differentiation, using $\Lambda = -F'' + i\{a,F'\} + aFa$, gives

$$\partial_{t_1}\partial_{t_2}B_a(t_1,t_2) = e^{iat_1}\Lambda(t_2-t_1)e^{-iat_2}. \tag{46}$$

Thus four endpoint derivatives recover two ordinary force kernels while retaining their common J. This does not model the forces themselves as independent probes.

A positive subtraction avoids differentiating pinching. Define the probability measure and its conditional function

$$\rho(dx,dy) = \mathbb{E}\,\mathrm{tr}_m[P_h(dx)V^\dagger P_K(dy)V], \qquad \mu(dx) = \rho(dx,\mathbb{R}), \tag{47}$$

$$\ell_s(x) = \mathbb{E}_\rho[(y-(1-p/m)x)^2 e^{isy} \mid x]. \tag{48}$$

By the spectral expansion of R, $\ell_s(h)$ is the exact annealed regression of $\Lambda(s)$ onto deterministic energy functions. The Hilbert-space projection identity and pinching contraction give

$$\mathcal{V}_{N,m}(s) = \mathbb{E}\,\mathrm{tr}_m\,\Lambda^\dagger\Lambda - \int |\ell_s|^2 d\mu \;\geq 0, \tag{49}$$

$$\|X_0(s)\|_2^2 \leq \sum_m b_m m^2 \mathcal{V}_{N,m}(s). \tag{50}$$

The difference is residual sample-dependent conserved variance; this bound is sufficient, not necessary. A uniform central-sector estimate $\mathcal{V}_{N,m} = O(p^{-4})$, with the controlled tails, would suffice. The established leading theorem yields only $\sum_m b_m \mathcal{V}_{N,m} = o(p^{-2})$. The rank–chaos estimate (41) does not by itself bound this charged, energy-regressed quantity.

8 Literature boundary, verification and next calculation

The finite-index trace rule in BNR [3], Eqs. (3.11)–(3.12), retains all intersections. Its subsequent leading double-scale simplification drops triple intersections. For a size-one charged probe, one marked site lies in two independent Hamiltonian index sets with probability $(p/N)^2 = \lambda_N/N$; two probe indices coincide with probability $1/N$. Both matter at connected order. The published neutral-probe four-point formula cannot simply be specialized to this charged, subleading problem. Endpoint derivatives and regression subtraction must be controlled together.

The distinction that motivated Sections 5–6 is explicit:

$$e^{isH} = \sum_{n\geq 0} \frac{(is)^n}{n!} \sum_{k=0}^{n} \mathsf{W}_k(H^n), \qquad :e^{isH}: = \sum_{k\geq 0} \frac{(is)^k}{k!} \mathsf{W}_k(H^k). \tag{51}$$

The terms with $k < n$ contain lower-chaos contractions, already included in (41). For example, $\mathsf{W}_1(H^3) = 2CH + \Phi(H)$ need not be a scalar multiple of H; a scalar Gaussian prefactor is not the argument used here. The remaining task is the coherent transfer between adjacent charge sectors in the actual force sandwich. It must connect (42) to the charged four-point and the energy regression (50); a Gaussian-chaos multiplier cannot simply be declared to commute with H. Every proposed limiting kernel must recover the earlier equal-time dynamic variance

$$W = 4J_s^4 \left[\frac{1}{\lambda} + \frac{4q}{1+q}\right], \qquad q = e^{-4\lambda}. \tag{52}$$

An independent-site approximation that retains only the first term misses the coherent common-coupling contribution.

Saved evidence. Exact rational Racah inversion and transition checks give 252 equalities over six finite systems and product lengths one through eight. The rational bounds are checked in 1800 parameter instances; four symbolic identities hold for general N, p, m. Six fixed-k symbolic limits check (22), whose general proof is the recurrence above. Independently constructed CAR channels in three systems reproduce the rank law through five factors. Nine complete Gaussian coefficient tensors through degree three verify the single-$k!$ bound. Eighteen complete Hermite coefficient cases through H^4 check creation commutators including lower chaoses; 43 exact one-step laws check the conditional bound and rank triangle. Three finite charged-probe systems verify the four-point second-moment sum and endpoint derivative identity. Five exact-formula evaluations at 80/120 digits agree within 10^{-60}; the large-k case checks numerical evaluation, not a fixed-k asymptotic. These are reproducibility checks, not substitutes for the uniform proof and not interval-certified rounding bounds.

References

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